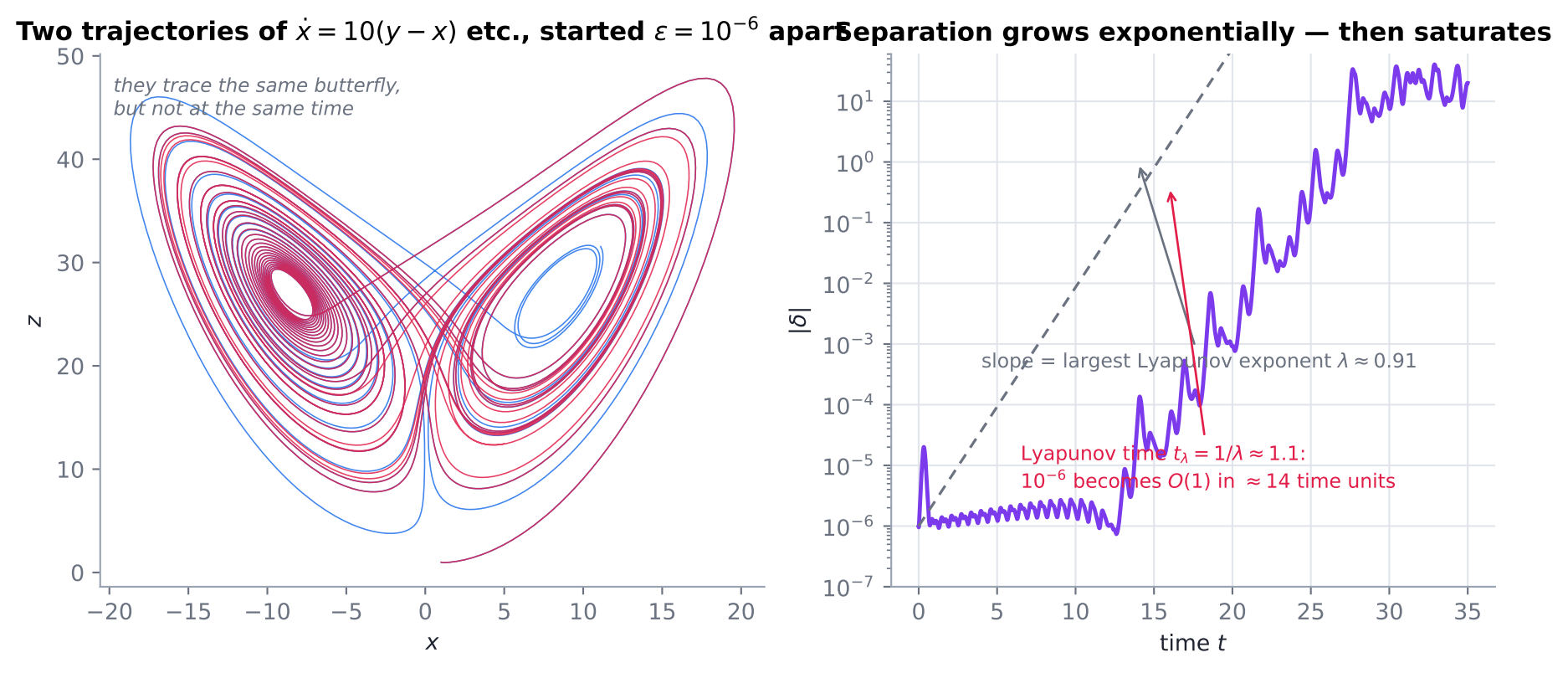

Two trajectories of $\dot{x} = 10(y - x)$ etc., started $\varepsilon = 10^{-6}$ apart
they trace the same butterfly,
but not at the same time
z
x
Separation grows exponentially — then saturates
$|\delta|$
time t
slope = largest Lyapunov exponent $\lambda \approx 0.91$
Lyapunov time $t_\lambda = 1/\lambda \approx 1.1$:
10^{-6} becomes $O(1)$ in ≈ 14 time units